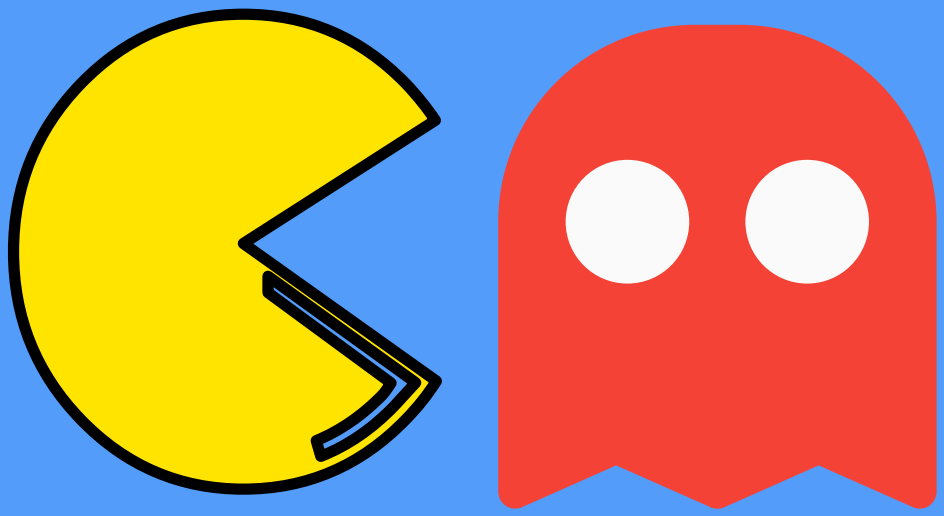
Arcade games

micro:bit Course

Module: Micro:bit Game Design

PACMAN



Lesson 4

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MICROBIT PACMAN

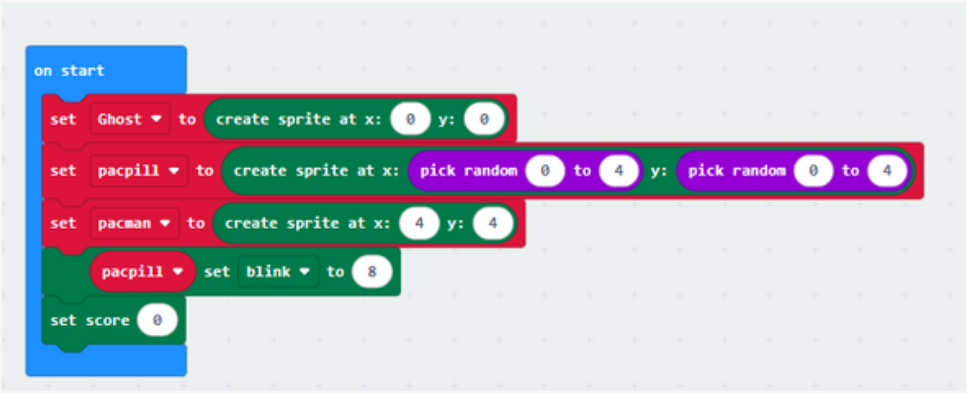
In this activity you will create a pacman game on the micro:bit. You will code pacman to be controlled by the accelerometer, so you will have to tilt your micro:bit to move pacman around Pacma

1



From Variables

2



From Games

From Math



When the micro:bit starts, create three sprites - pacman, a flashing pacpill and a ghost Number 1). Use random to place the pacpill randomly. Create a score for your game (number 2).

Further Information

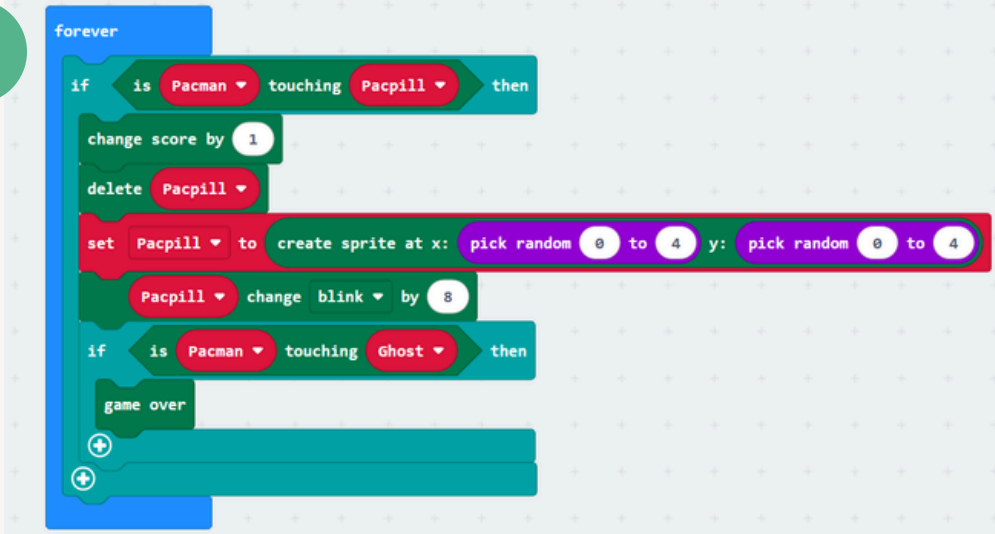
EXAMPLE

Click here for the micro:bit foundation micro:bit tutorial

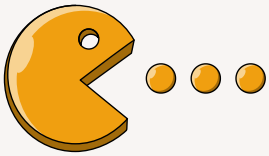
Click here for example code

Logic

3



This endless cycle is quite simple:
If Pac-Man touches a Pac-Pill, the score increases by 1 and the Pac-Pill disappears, only to reappear in a random position near the centre
If (on the other hand) Pac-Man touches a ghost, the game ends



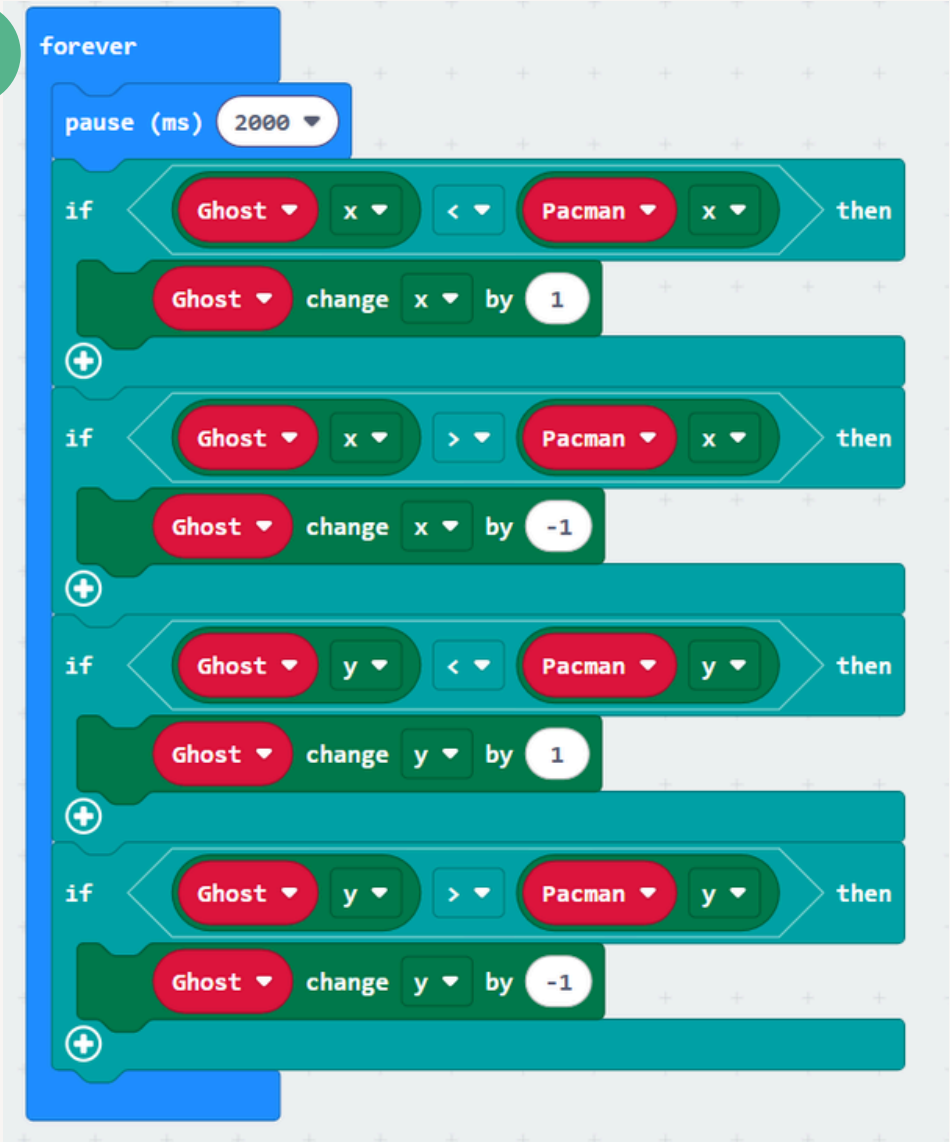
4

Input



But how do we make our Pac-Man move? We use the accelerometer built into our Micro:bit.
This forever loop controls pacman by tilting the micro:bit. Feel free to tune these numbers! I have found they work well, but you may be able to fine tune this game further!

5



Add another forever loop for the ghost to chase pacman. The time delay is important - if you change the time delay in the forever loop the ghost will speed up or slow down when chasing pacman.



1) Make the ghost move faster as the score increases.

2) Add sound effects and background music to make the game engaging.

3) Display a startup logo to show it's Pac-Man on the micro:bit!